
A Watermarking with Two Signatures

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Outline

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Introduction

- Most of the recent works in watermark can be grouped into two categories : frequency domain methods and spatial domain
- Watermarking in frequency domain firstly transforms data into frequency domain using DCT or FFT
- Watermarking in spatial domain adds the watermark without any transformation

Introduction(cont.)

- This paper provides two different extraction methods in the frequency domain
- One requires source image to extract the watermark , while the other doesn't
- The watermarked image is non-invertible, and hence is robust to the IBM attack
- this watermarking scheme provides more robustness against the invertible watermarking problem in watermarking

Insertion of Watermark

- The proposed watermarking has two methods of watermark insertion
- In the first insertion, two frequencies are chosen by random numbers and their values are slightly changed
- In the second insertion, the values of the same frequencies are modified

First Insertion of Watermark

- First, the source image is transformed to frequency domain by a 2-D FFT
- When a frequency is selected as a candidate site, the neighboring frequency along the x or y-axis is selected also
- Therefore four coefficients are modified at a time

First Insertion of Watermark(cont.)

- The watermark consists of a sequence of real numbers $X = x_0, x_1, \dots, x_N$ and x_i is chosen randomly according to $N(0,1)$
- Let $V = v_0, v_1, \dots, v_N$ be a sequence of coefficients, where even v_{2n} and v_{2n+1} are real and imaginary coefficients of a frequency
- The insertion formula is given by :

$$v_i' = v_i \times (1 + \alpha x_i)$$

First Insertion of Watermark(cont.)

- The scheme of the first insertion of watermark is similar to the frequency domain based method
- These four coefficients will be also used at the second watermark insertion

Second Insertion of Watermark

- The proposed method inserts the second watermark into sequence of values $V' = v'_0, v'_1, \dots, v'_N$ which is modified by the first watermark
- The modified coefficients of A_l and B_l during the first watermark are denoted as $A'_{lr}, A'_{li}, B'_{lr}, B'_{li}$

Second Insertion of Watermark(cont.)

- The magnitudes of A_l and B_l after the first watermark insertion are denoted as $F(A_l)$ and $F(B_l)$ where

$$F(A_l) = \sqrt{A_{lr}'^2 + A_{li}'^2}$$

$$F(B_l) = \sqrt{B_{lr}'^2 + B_{li}'^2}$$

Second Insertion of Watermark(cont.)

- We compare $F(A_l)$ and $F(B_l)$ by a randomly generated key sequence, $K = k_0, k_1, \dots, k_M$
- If k_l is zero, it sets $F(A_l) > F(B_l)$
- If $F(A_l)$ is smaller than $F(B_l)$ the coefficients of the frequencies (A'_{lr}, A'_{li}) and (B'_{lr}, B'_{li}) are swapped each other
- If k_l is one, it sets $F(A_l) < F(B_l)$

Second Insertion of Watermark(cont.)

- The second insertion of watermark scheme does not modify coefficient values but only swaps the corresponding pair coefficients
- Each key k_l is inserted N/M times so that key extraction can be robust to the various noise and corruption
- In order to keep the image not degraded, neighboring frequencies are chosen for the watermark

Extraction of Watermark

- The proposed watermarking scheme extracts the second watermark first
- If we can't get rightful ownership by second watermark, then we extract the first watermark to get certain rightful ownership

Extraction of Watermark(cont.)

- The extraction sequence is $K^* = k_0^*, k_1^*, \dots, k_M^*$ as follows :

$$\begin{array}{ll} \text{if } F^*(A_l) > F^*(B_l) & k_n^* = k_n^* - 1 \\ \text{else} & k_n^* = k_n^* + 1 \end{array}$$

where k_n^* , $n=0\dots M$, is initially set to zero

Extraction of Watermark(cont.)

- The similarity measure of the second watermark is calculated by the following formula.

$$S(K, K^*) = \sum_{n=1, M} (2k_n - 1)k_n^* / N$$

- If there is an exact match between the inserted key sequence and the exacted one, then $S(K, K^*)$ will have the value of 1

Extraction of Watermark(cont.)

- The extraction formula for the first watermark is $x_i^* = (v_i^* / v_i - 1) / \alpha$
- the similarity measure of the first watermark is calculated by following formula,

$$S(X, X^*) = \frac{\sum_{i=1, M} x_i x_i^*}{\sum_{i=1, M} x_i \sum_{i=1, M} x_i^*}$$

The IBM Attack

- This research was supported at 1996
- The concept is described as follows :

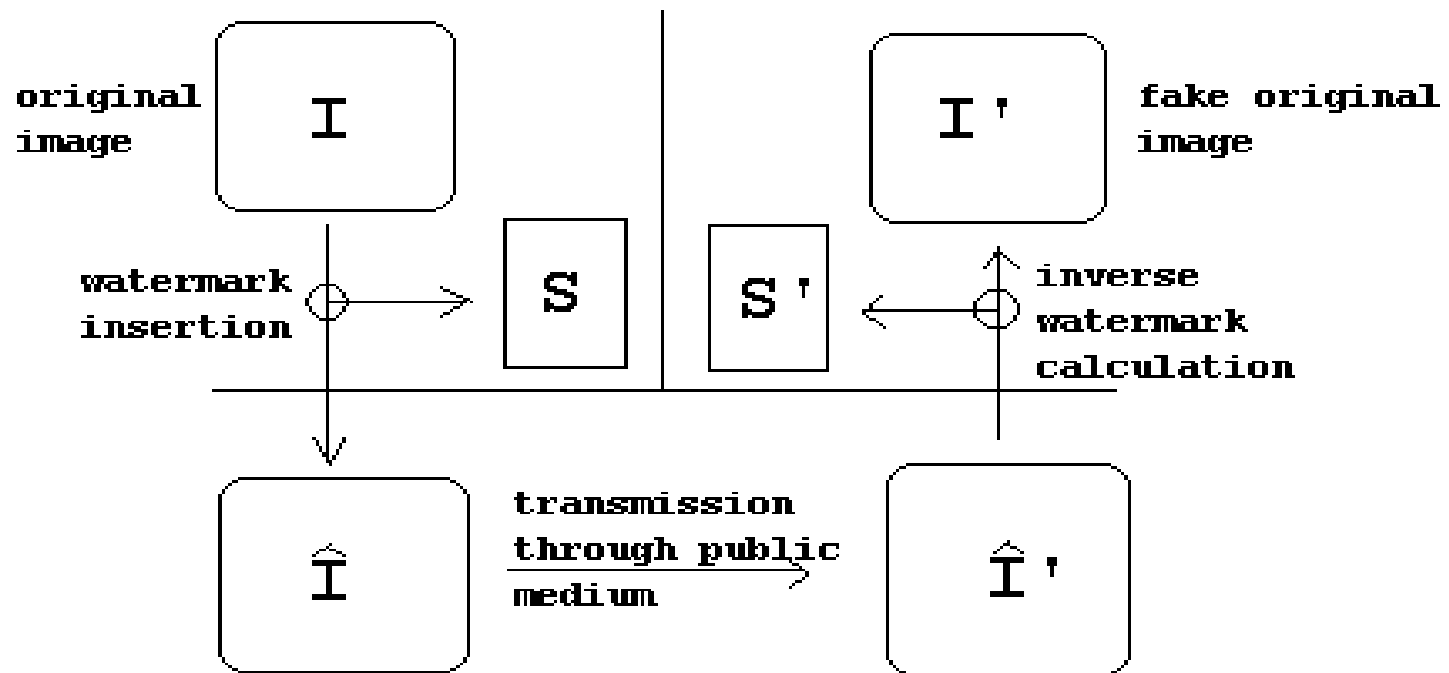
$$D' = D + X_1$$

$$D_F = D' - X_2$$

$$D = D_F + X_2$$

where D is a source document and D' is a watermarked document. D_F is known as the counterfeit original

The IBM Attack(cont.)



The IBM Attack(cont.)

- The class of watermarking scheme that can be attacked by creating a “counterfeit original” as the *invertible* watermark schemes
- The first method adds an invertible watermark, while the second method adds a non-invertible one
- So the proposed watermarking scheme is robust to the invertible attack.

Experimental Result

- Experiments are conducted on the 256*256 source image (show below)
- The generated key sequence, is 0000010111001010.

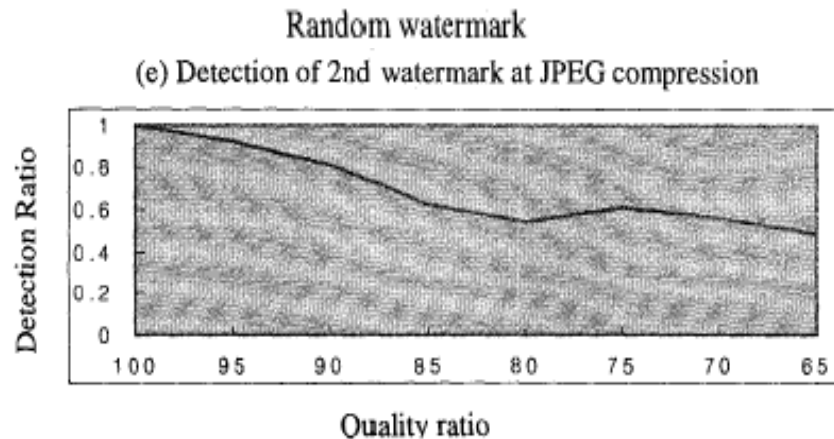
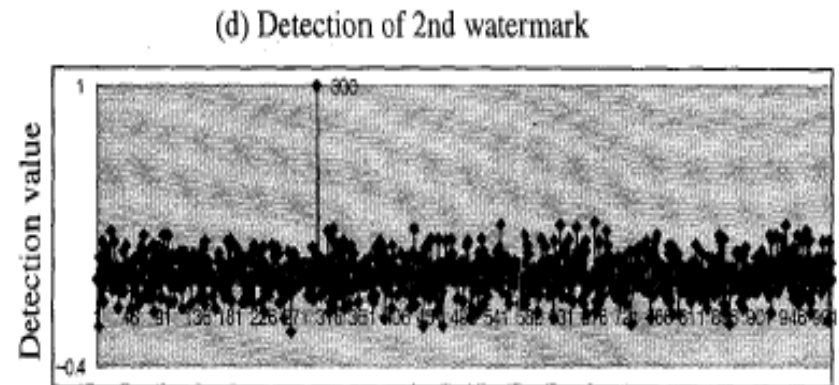
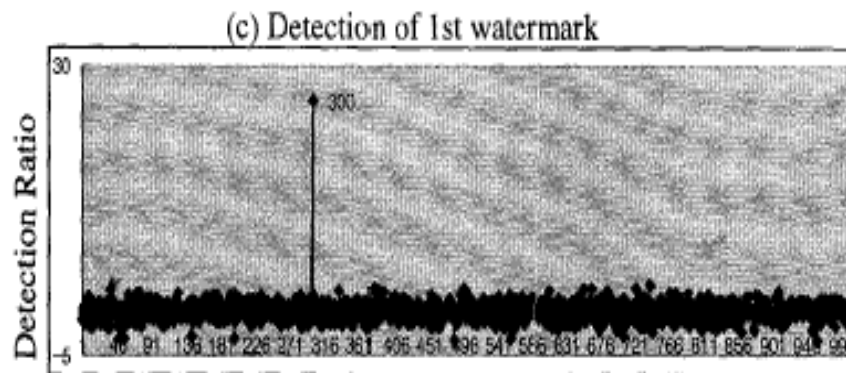


(a) Source image



(b) Watermarked image

Experimental Result



<c> first watermark detection

<d>the detection of the second watermark

<e>the robustness of the technique to JPEB compression

Figure 2: Source, Watermarked image and Simulation result

Conclusion

- The proposed watermarking scheme is shown to be robust to the IBM attack as it does not need the source image to extract the second watermark